

Alimony Rights and Young Women's Education in Brazil

Tuffy Licciardi Issa

2025-10-25

This document is a minimal, reproducible sample of my empirical workflow.
Full code and additional projects: github.com/tuffyli

Objectives

This sample studies whether the expansion of alimony rights in Brazil affected the educational attainment of young women who became newly eligible for those rights. The exercise is inspired by Rangel (2006), with an additional propensity score matching step used to improve balance between treated and control women before estimating treatment effects.

Data

The analysis uses PNAD microdata for 1992, 1993, and 1995. The full PNAD files are not redistributed because of size constraints, but the repository includes a filtered dataset with the variables used in this exercise.

```
data <- readRDS(project_path("data", "final_filtered_9295.rds"))
summary_table <- readRDS(project_path("data", "summary.rds"))

knitr::kable(
  summary_table,
  digits = 2,
  caption = "Weighted descriptive statistics for the analysis sample"
)
```

Table 1: Weighted descriptive statistics for the analysis sample

	Mean	SD	Min	Max
Treatment	0.45	0.50	0	1
Female = 1	1.00	0.00	1	1
Race (White or Asian = 1)	0.55	0.50	0	1
Highest education	4.22	0.56	0	9
Age	21.04	2.41	15	24
Years of education	6.61	3.37	1	17
Last grade completed	4.47	2.00	1	9
Course enrollment	2.48	1.65	0	9
Household per-capita income	173221.66	269756.48	0	9549387

	Mean	SD	Min	Max
Pension/alimony receipt = 1	0.00	0.05	0	1
CBO group	4.34	2.58	0	8
Male children in household	0.57	0.72	0	12
Female children in household	0.55	0.71	0	6
Total children	1.17	1.00	0	13

Treated individuals are women aged 15-24 in non-civil unions, either consensual or religious-only, who are household heads or spouses. Controls are otherwise similar women in legally recognized unions, either civil or civil plus religious.

Propensity Score Matching

I estimate propensity scores separately by year and match treated women to controls using nearest-neighbor matching with replacement. The balance plots below compare standardized mean differences before and after matching.

```
match_year <- function(year, input_data) {
  df_psm <- input_data %>%
    filter(ano == year) %>%
    select(
      treatment, uf, age, cor, anos_estudo, peso_pessoa,
      ultima_serie_concluida, tipo_curso_frequenta, grupo_cbo,
      trabalhou_ultimo_ano, renda_dom_per_capita,
      dummy_filhos_homens_dom, dummy_filhos_mulheres_dom,
      pensao_dummy, total_filhos
    ) %>%
    mutate(uf = as.factor(uf)) %>%
    filter(
      if_all(
        c(
          treatment, uf, cor, peso_pessoa, dummy_filhos_homens_dom,
          dummy_filhos_mulheres_dom, renda_dom_per_capita, total_filhos
        ),
        ~ !is.na(.x)
      )
    )

  match_model <- matchit(
    treatment ~ age + cor + uf + dummy_filhos_homens_dom +
      dummy_filhos_mulheres_dom + renda_dom_per_capita + total_filhos,
    data = df_psm,
    s.weights = df_psm$peso_pessoa,
    caliper = 0.05,
    method = "nearest",
    replace = TRUE
  )

  match_summary <- as.data.frame(summary(match_model)$nn)
  match_summary_path <- project_path("match-tables", paste0(year, "_match_summary.tex"))
  dir.create(dirname(match_summary_path), showWarnings = FALSE, recursive = TRUE)
```

```

writeLines(
  kable(
    match_summary,
    format = "latex",
    booktabs = TRUE,
    caption = paste("Matching sample counts,", year),
    align = "lccc"
  ),
  match_summary_path
)

custom_labels <- c(
  distance_weighted = "Distance (weighted)",
  uf = "UF",
  cor = "Race/Color",
  dummy_filhos_homens_dom = "Male children in household",
  dummy_filhos_mulheres_dom = "Female children in household",
  age = "Age",
  renda_dom_per_capita = "Household per-capita income",
  total_filhos = "Total children"
)

balance_plot <- love.plot(
  match_model,
  stats = "mean.diffs",
  abs = TRUE,
  threshold = 0.10,
  colors = c("black", "red"),
  line = TRUE,
  var.names = custom_labels,
  drop.distance = TRUE
) +
  labs(title = paste("Covariate balance:", year)) +
  theme_minimal(base_size = 5) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "bottom"
  )

list(
  matched_data = match.data(match_model) %>% mutate(ano = year),
  balance_plot = balance_plot
)
}

match_results <- set_names(c(1992, 1993, 1995)) %>%
  map(match_year, input_data = data)

plots <- map(match_results, "balance_plot")

top <- cowplot::plot_grid(
  plots[["1992"]],
  plots[["1993"]],

```

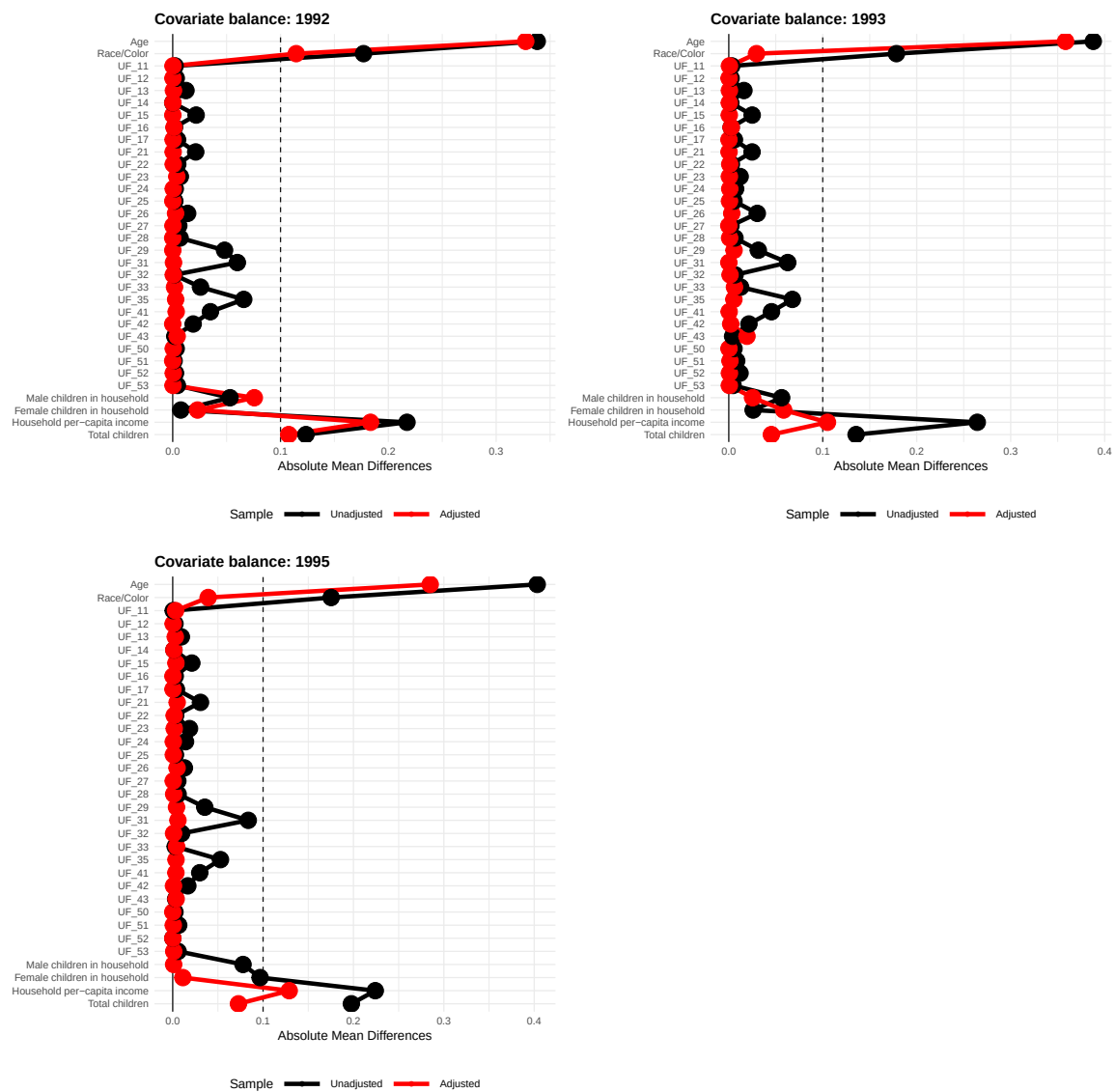
```

ncol = 2,
align = "hv",
rel_widths = c(1, 1)
)

bottom <- cowplot::plot_grid(
  plots[["1995"]],
  ncol = 2,
  align = "hv",
  rel_widths = c(1, 1)
)

cowplot::plot_grid(top, bottom, ncol = 1, rel_heights = c(1, 1.05))

```



Regression

To assess treatment effects, I estimate progressively richer weighted OLS specifications on the matched sample: a baseline model, a model with UF-by-year fixed effects, and a fixed-effect model with additional controls.

```
df_psm <- match_results %>%
  map("matched_data") %>%
  bind_rows() %>%
  mutate(
    ano = relevel(factor(ano), ref = "1992"),
    uf_ano = interaction(uf, ano, sep = "_")
  )

model_nc <- feols(
  anos_estudo ~ treatment + i(ano, treatment, ref = "1992"),
  data = df_psm,
  weights = ~weights,
  vcov = "hetero"
)

model_fe <- feols(
  anos_estudo ~ treatment + i(ano, treatment, ref = "1992") | uf_ano,
  data = df_psm,
  weights = ~weights,
  vcov = "hetero"
)

model_cc <- feols(
  anos_estudo ~ treatment + i(ano, treatment, ref = "1992") +
    total_filhos + cor + age | uf_ano,
  data = df_psm,
  weights = ~weights,
  vcov = "hetero"
)

fixest::etable(
  list("No FE" = model_nc, "UF x Year FE" = model_fe, "FE + Controls" = model_cc),
  tex = TRUE,
  drop = "Constant",
  dict = c(
    "anos_estudo" = "Years of education",
    "treatment$" = "Treatment (1992)",
    "i\\(ano, treatment.*\\)::1993" = "Treatment x 1993",
    "i\\(ano, treatment.*\\)::1995" = "Treatment x 1995",
    "age" = "Age",
    "cor" = "Race/Color",
    "total_filhos" = "Total children",
    "treatment x ano = 1993" = "Treatment x 1993",
    "treatment x ano = 1995" = "Treatment x 1995"
  ),
  se.below = TRUE,
  signif.code = "letters",
```

```

digits = 3,
fitstat = c("n", "r2", "rmse"),
title = "Treatment Effects by Year (reference year = 1992)",
notes = "SEs are heteroskedasticity-robust. Regression weights are inherited from the matched sample."
)

```

Table 2: Treatment Effects by Year (reference year = 1992)

Dependent Variable: Model:	Years of education		
	(1)	(2)	(3)
<i>Variables</i>			
treatment	-1.77 ^a (0.077)	-1.66 ^a (0.100)	-1.25 ^a (0.095)
treatment × ano = 1993	0.307 ^a (0.080)	0.230 (0.146)	0.112 (0.136)
treatment × ano = 1995	0.452 ^a (0.077)	0.176 (0.147)	0.055 (0.137)
Total children			-0.915 ^a (0.028)
Race/Color			0.803 ^a (0.057)
Age			0.260 ^a (0.011)
<i>Fixed-effects</i>			
uf_ano		Yes	Yes
<i>Fit statistics</i>			
Observations	17,779	17,779	17,779
R ²	0.05238	0.11600	0.21156
RMSE	66.252	63.990	60.432

Heteroskedasticity-robust standard-errors in parentheses

Signif. Codes: a: 0.01, b: 0.05, c: 0.1

SEs are heteroskedasticity-robust. Regression weights are inherited from the matched sample.

The estimates do not provide evidence of a statistically significant effect of alimony eligibility on young women's years of education. Because the models explain only a small share of variation in schooling outcomes, the results should be interpreted as a compact empirical exercise rather than a definitive causal estimate.